# *END OF CHEAP SOVEREIGNTY REPORT*

***An examination of how the end of ultra-low interest rates is reshaping sovereign debt, fiscal sustainability, and global capital markets.***

IFSA

NIT Trichy

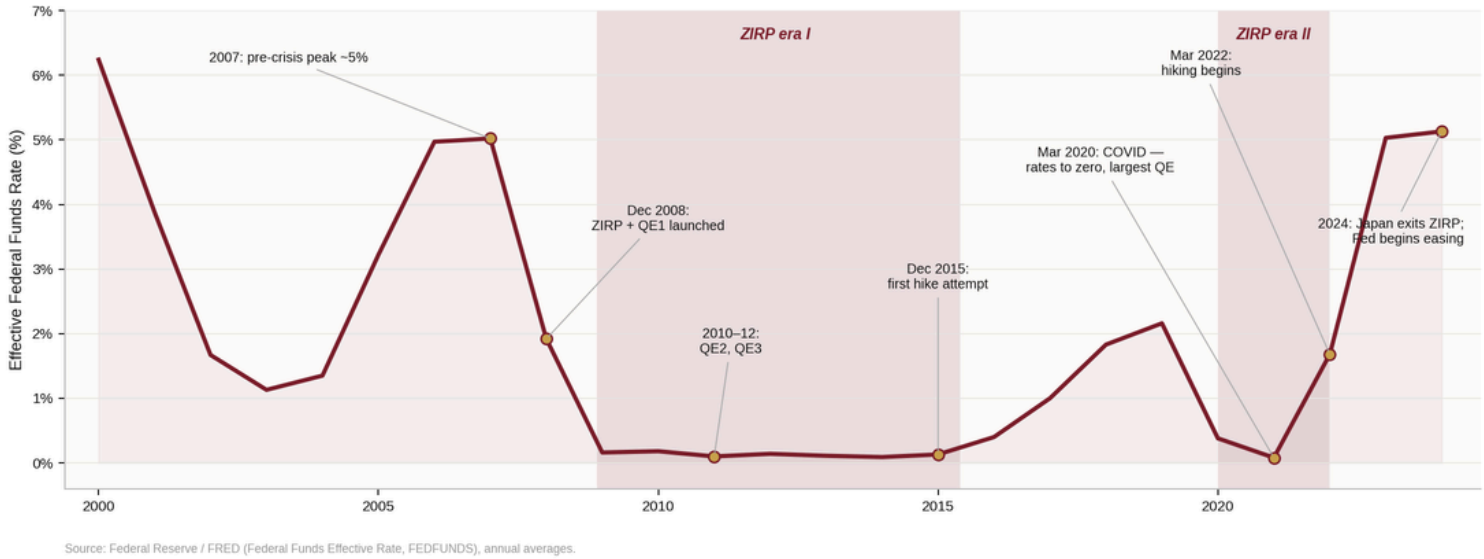
*PREPARED BY IFSA, NIT Trichy*

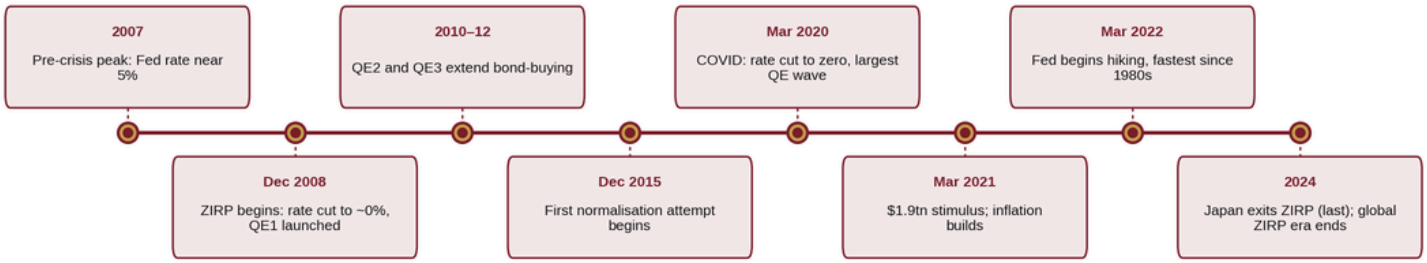
# The Rise and Fall of ZIRP (2008-2024)

***THE FEDERAL FUNDS RATE AND KEY POLICY MILESTONES, 2000–2025***

For more than a decade, **ultra-low interest rates allowed governments to expand borrowing with limited fiscal consequences.** That regime has fundamentally changed. Persistent inflation, geopolitical fragmentation, demographic shifts, and rising refinancing costs have ended the era of cheap sovereign financing. This report examines the structural forces driving this transition, assesses the risks across major economies, and explores what a **higher-for-longer world means for sovereign debt, capital markets, and long-term fiscal sustainability.**

The federal funds rate, the Federal Reserve's benchmark policy rate, sets the foundational cost of borrowing across the US economy. Its path over ***twenty-five years*** provides the clearest single record of the cheap-money era, traced through the marked milestones below:



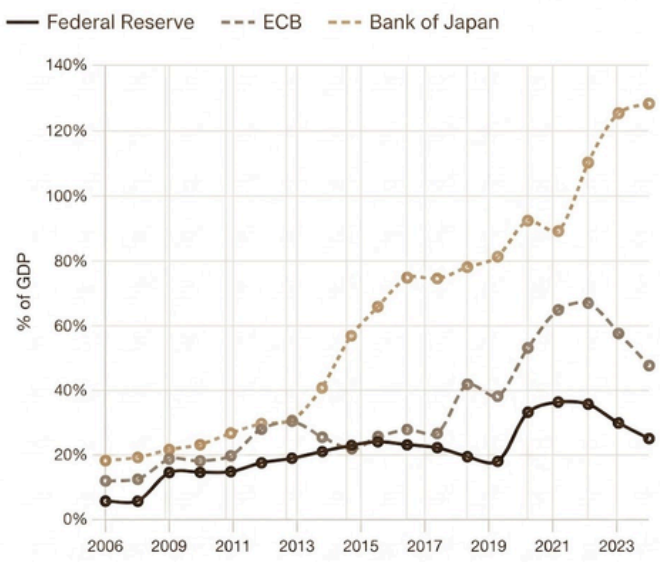

## *The Big Picture*

The significance lies not in any rate level but in the contrast: a prolonged period at the floor against the speed of the ascent. Governments built their **fiscal systems around borrowing costs that seemed permanent**, then had little time to adjust when they ended.

# The Footprint of Quantitative Easing

***ULTRA-LOW RATES AND QUANTITATIVE EASING FUNDAMENTALLY ALTERED SOVEREIGN FINANCING CONDITIONS.***

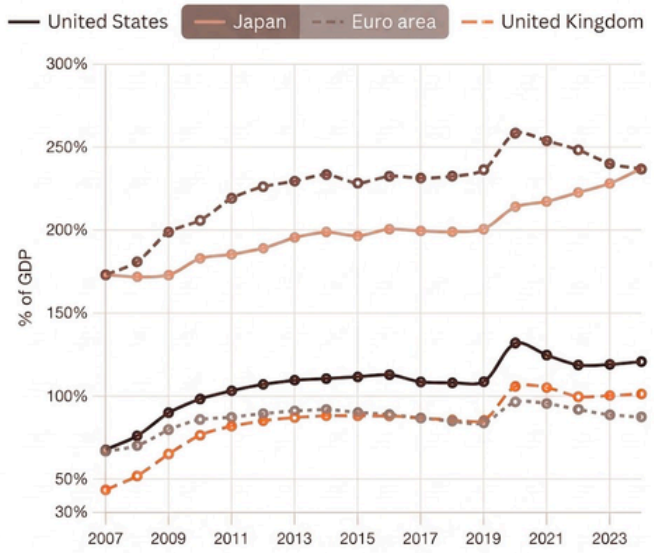
## *Central Bank Balance Sheet Expansion*




***Extraordinary monetary easing permanently expanded central bank balance sheets.***

Following the Global Financial Crisis and COVID-19, major central banks expanded their balance sheets to unprecedented levels. The Federal Reserve's assets peaked at **~37% of GDP, while the Bank of Japan exceeded 125%**, becoming one of the largest holders of sovereign debt. Although quantitative tightening has reduced holdings, balance sheets remain structurally larger than before 2008, making central banks a permanent pillar of government financing rather than temporary crisis responders.

## *The Rise of Sovereign Debt*




***Years of low borrowing costs enabled a sustained expansion in government debt.***

Years of ultra-low interest rates enabled governments to accumulate record levels of debt with limited fiscal pressure. Advanced economy debt increased from **~74% of GDP in 2007 to over 112% by 2024**, while Japan's debt approached **240% of GDP.**

As ZIRP-era bonds mature and r**efinancing occurs at 4–5% yields,** debt servicing not debt accumulation has become the primary fiscal challenge.

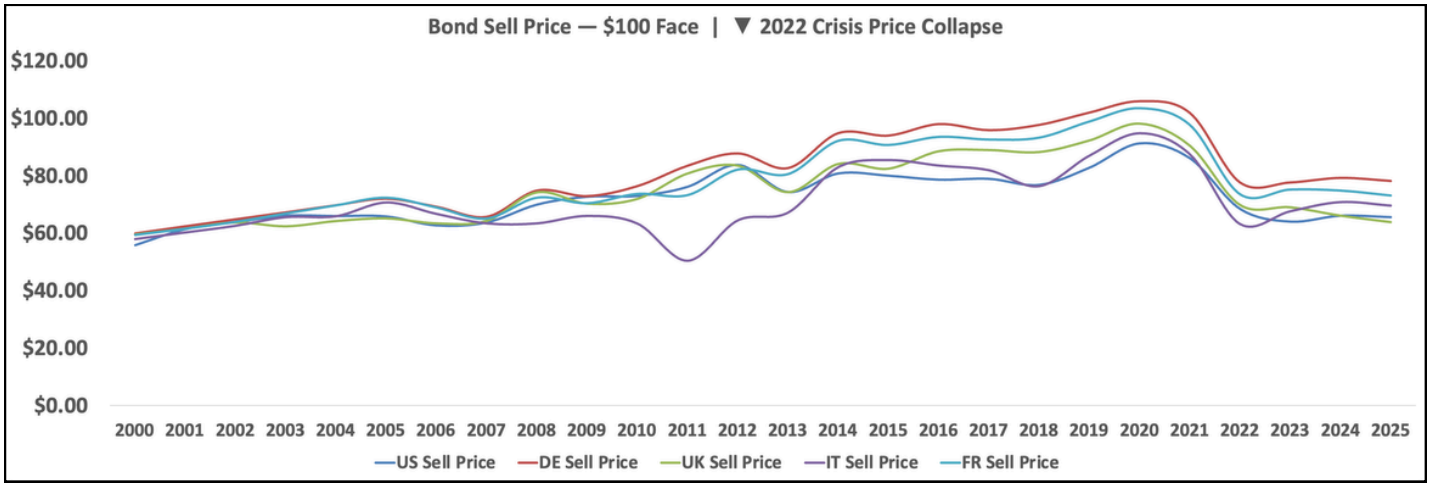
Together, these trends illustrate the defining legacy of the cheap-money era: central bank intervention enabled an unprecedented expansion in sovereign borrowing, but the return of higher interest rates has fundamentally altered its sustainability.

Governments now face a **structural shift from managing debt accumulation to managing refinancing costs, making fiscal resilience increasingly dependent on interest rates rather than debt levels alone.**

# The Return of Bond Vigilantes

***HOW INFLATION, DEFICITS, AND HIGHER YIELDS ARE RESHAPING GOVERNMENT BORROWING.***

## *From the 2022 Global Crash to the Higher-for-Longer Regime*




## *Interest Rates Through the Monetary Cycle*

Three major central banks shaped the global rate environment over 25 years. The Federal Reserve, ECB, and Bank of Japan each executed distinct variants of ZIRP, creating a G7 synchronised ultra-low rate environment that dramatically suppressed sovereign borrowing costs and then reversed with historic speed.

**The Federal Reserve executed the sharpest tightening cycle in four decades, raising the effective funds rate from 0.08% in 2020 to 5.33% by July 2023 (+525bps in under 18 months).** The ECB followed, delivering 450bps of hikes between July 2022 and September 2023 from a negative rate base **of -0.50%**. The Bank of Japan remained the last holdout, finally beginning its exit from negative rates and Yield Curve Control in 2024.

## *The Repricing of Fixed Income*

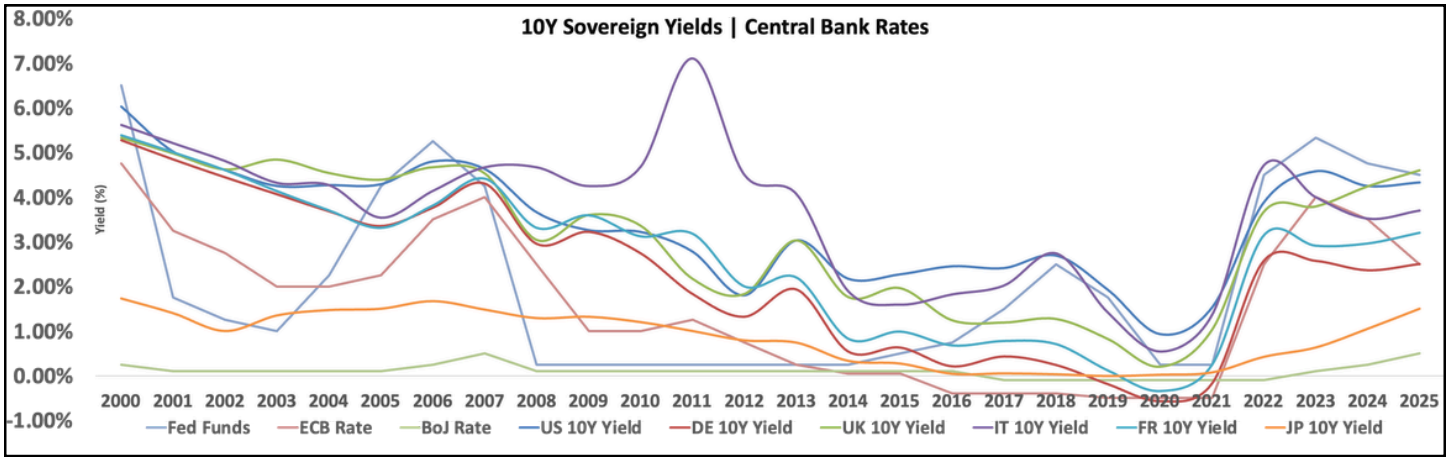
The inflation shock that broke the ZIRP regime was extraordinary. US CPI surged from **1.4% (Jan 2021) to 9.1% in June 2022,** the highest since November 1981. **Eurozone HICP peaked at 10.6% in October 2022.** The shock reflected simultaneous demand-side (fiscal transfers, reopening) and supply-side pressures that central banks were illpositioned to anticipate
The 2022 global bond selloff was, by every major metric, the worst in modern financial history.

The Bloomberg Global Aggregate Bond Index lost approximately **16.0% in total return terms, exceeding losses in any prior calendar year since the index's 1990 inception.**

# The Return of Bond Vigilantes

***HOW INFLATION, DEFICITS, AND HIGHER YIELDS ARE RESHAPING GOVERNMENT BORROWING.***

## *From the 2022 Global Crash to the Higher-for-Longer Regime*




## *Sovereign Debt Sustainability: Debt-to-GDP & Interest Burden*

The repricing of sovereign bond markets transmits into higher borrowing costs **with a 2 to 5 year lag, as existing low rate debt matures and must be refinanced at prevailing market rates.** This creates the "refinancing cliff" with large volumes of cheap ZIRP era issuance maturing and requiring replacement at **4% to 5% rates.**

The fundamental sustainability equation is **$\Delta d = (r - g) \times d - pb$.** When $r > g$, as is now the case across most G7 economies, debt ratios rise automatically without new spending. This structural shift from the ZIRP era $r < g$ condition represents the most consequential change in sovereign debt dynamics in a generation.

## *US 10-Year Yield & The Federal Reserve Balance Sheet*

The US 10-year Treasury yield is the world's most important risk-free benchmark. **The October 2023 peak of 5.02%** the highest since 2007 was not solely driven by near-term Fed Funds expectations, but by the re-emergence of the term premium: investor compensation for fiscal risk, inflation uncertainty, and the quantitative tightening supply overhang.

The Fed balance sheet expanded from **$1T (pre-GFC) to a peak of $8.96T in April 2022 ($4.7T during COVID alone).** Quantitative tightening, running at **$95B/month,** has reduced holdings to **$7T** but the stock of duration supply already absorbed by markets remains a permanent headwind to long-end yields.

# The Return of Bond Vigilantes

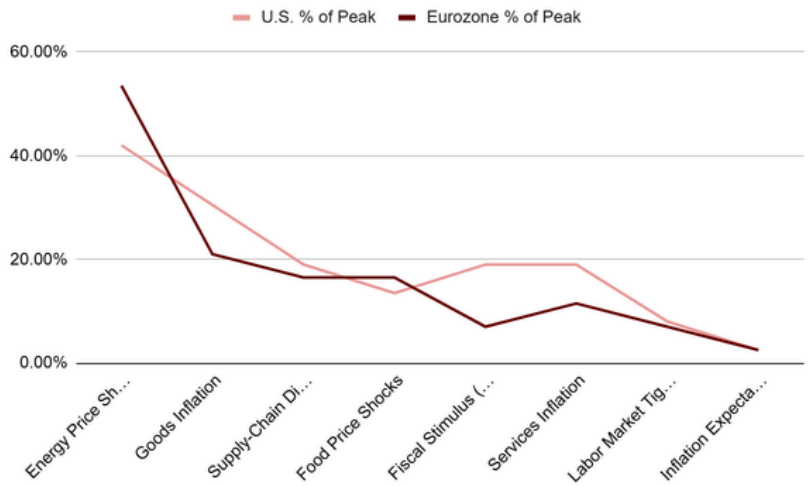
***HOW THE 2022 GLOBAL SELLOFF ENDED THE ERA OF CHEAP MONEY***

## *The Great Bond Repricing*

The collapse of the zero-rate regime fundamentally altered global fixed-income markets. Persistent inflation, aggressive monetary tightening, quantitative tightening, and geopolitical supply shocks combined to produce the worst annual bond market performance on record. Beyond the immediate losses, the repricing established a new investment environment in which higher yields, greater duration risk, and **increased refinancing costs** are likely to **remain structural features rather than temporary disruptions.**

From 2020 to 2024, a series of overlapping shocks, pandemic disruptions, unprecedented fiscal and monetary stimulus, supply-chain breakdowns, labor shortages, and the Russia–Ukraine energy shock, drove inflation to 40-year highs across the globe. **As inflation became persistent and broad-based, central banks were forced to end the era of near-zero interest rates. The return of meaningful borrowing costs has fundamentally altered sovereign debt dynamics.**

## *The Inflation Shock Explained*




Energy prices and goods inflation were the largest contributors to peak inflation in both the United States and the Eurozone.

***Supply-chain disruptions, food-price pressures, labor market tightness, services inflation***, and fiscal stimulus further amplified price pressures.

Rather than acting independently, these forces reinforced one another, creating a synchronized inflation cycle that became the most significant inflation shock in more than four decades.

The inflation shock forced central banks to abandon the near-zero interest rate policies that had prevailed since 2008. As inflation peaked at **9.1% in the United States, 10.7% in the Eurozone, and 11.1% in the United Kingdom**, the Federal Reserve, ECB, and Bank of England launched one of the most aggressive tightening cycles in modern history.

# The Refinancing Challenge

***HIGHER INTEREST RATES HAVE FUNDAMENTALLY SHIFTED SOVEREIGN DEBT RISK FROM DEBT ACCUMULATION TO DEBT SERVICING.***

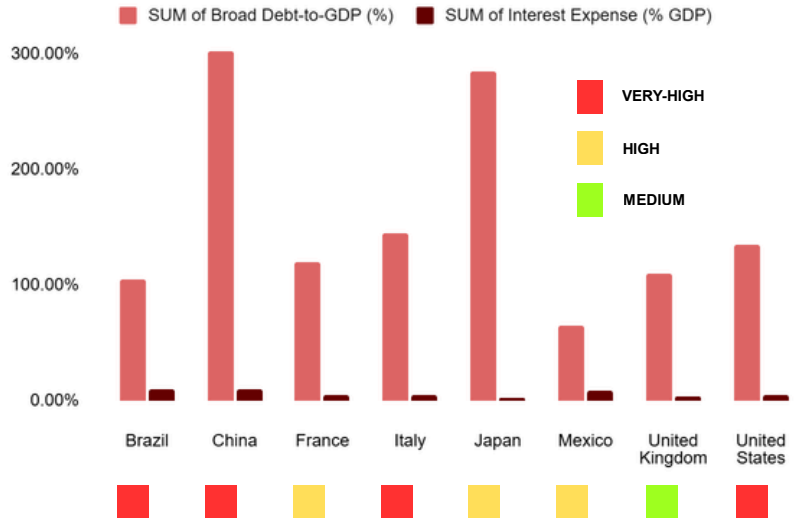
## *The Great Bond Repricing*

The end of the ZIRP era transformed sovereign debt from a manageable liability into a growing fiscal constraint. Governments accumulated large debt burdens during a decade of **near-zero borrowing costs, but refinancing that debt has become significantly more expensive as interest rates moved higher.**

China and Japan now carry broad debt burdens of approximately **302% and 285% of GDP, while Brazil's interest expense has risen to nearly 9.5% of GDP.**

The United States combines a broad debt burden of **135% of GDP** with rising servicing costs, increasing exposure to future refinancing pressure. As a result, sovereign vulnerability is increasingly determined not only by how much debt exists, but by the cost of rolling it over.

## *The Debt Servicing Challenge*




The sovereign debt challenge did not emerge when governments accumulated debt.

It emerged when inflation ended the era of cheap money and **refinancing costs began to rise.**

As sovereign debt accumulated during the era of **near-zero interest rates, refinancing costs remained historically low.** That dynamic has now reversed.

With borrowing costs **normalizing, debt sustainability** is increasingly determined not by the size of public debt, but by the cost of servicing and refinancing it, exposing **highly indebted economies to growing fiscal pressure.**

# Why Rates will stay Higher?

***SEVEN STRUCTURAL DRIVERS OF PERSISTENT INFLATION AND HIGHER SOVEREIGN YIELDS***

From 2020, the global price level broke a **40-year disinflationary trend**, driven by structural, not cyclical, forces. **Seven long-lived drivers, operating together across three inflation channels, each lift the neutral rate and the cost of carrying sovereign debt.**

## *01 · DE-GLOBALIZATION*

**Time Horizon:** Permanent

**Mechanism:** The shift from efficient global supply chains to resilient local ones means goods cost more to produce. Just-in-time inventory is being replaced by just-in-case stockpiling, and production is moving closer to consumption - all of which carries a permanent cost premium.

**Impact:** **1-2%** permanent cost premium on goods; tariff pass-through adds **+0.3-0.5**pts to core CPI.

**Consequence:** A permanently higher inflation floor raises the neutral rate, meaning every government debt refinancing happens at structurally higher yields.

## *02 · DEFENCE SPENDING BOOM*

**Time Horizon:** 10 - 20 yrs

**Mechanism:** NATO's 2% floor is now a political minimum. Germany, Poland, Japan, and others are all rearming. Defence procurement is price-insensitive and crowds out civilian capacity.

**Impact:** **+$350-400bn** in additional annual NATO spending; Poland at **4.48%**, US at **3.5%**; NATO targeting 5% by 2035.

**Consequence:** Treaty-bound deficit expansion is non-discretionary - governments can't cut defence to balance budgets, structurally widening sovereign risk premia.

## *03 · US–CHINA DECOUPLING*

**Horizon:** 10 - 20 yrs

**Mechanism:** Export controls, tariff escalation, and investment screening are dismantling the $500bn+ annual flow of cheap Chinese goods -a bipartisan, structural shift, not a diplomatic tactic.

**Impact:** Domestic semiconductors cost **40–50%** more; reshored pharma runs **20–30%** pricier; effective tariffs exceed 30% across many categories.

**Consequence:** Term premium re-emerges on Treasuries; Fed credibility impaired; elevated real yields compress debt affordability.

## *04 · RESHORING & FRIEND-SHORING*

**Time Horizon:** 10 - 15 yrs

**Mechanism:** CHIPS Act, IRA, and IIJA direct $2T+ into domestic production; friend-shoring to Mexico, India, Vietnam, and Poland carries a 10-20% cost premium. Supply chains are being rebuilt along geopolitical lines rather than cost efficiency.

**Impact:** Near-shore wages are up **35-45%** since 2022; US semiconductor fabs run **50-100%** over budget; the IRA net cost alone is $400-800bn per decade; capex stays elevated for 10-15 years.

**Consequence:** Subsidy commitments widen structural deficits and surge bond issuance, eroding bond-market demand for real yield.

# Why Rates will stay Higher?

***HEATMAP OF INFLATIONARY FORCES BY COUNTRY***

| Structural Driver | USA | EU | UK | JAPAN | CHINA | EM Avg |
|---|---|---|---|---|---|---|
| De-Globalization | VERY HIGH | VERY HIGH | HIGH | HIGH | VERY HIGH | MED |
| US-China Decoupling | VERY HIGH | HIGH | HIGH | VERY HIGH | VERY HIGH | HIGH |
| Reshoring / FriendShore | VERY HIGH | VERY HIGH | HIGH | MED | MED | LOW |
| Defense Spending | HIGH | VERY HIGH | HIGH | HIGH | VERY HIGH | MED |
| Climate Transition CapEx | VERY HIGH | VERY HIGH | VERY HIGH | VERY HIGH | HIGH | MED |
| Aging Populations | MED | VERY HIGH | HIGH | VERY HIGH | VERY HIGH | LOW |
| Fiscal Dominance | VERY HIGH | HIGH | VERY HIGH | VERY HIGH | MED | HIGH |

## Heatmap Analysis

**United States:** Faces highest structural inflation risk as primary driver of deglobalization and US-China decoupling. Although domestic energy softens climate-CapEx pain, labour tightening is severe. Combined with large ficsal deficits and rising debt, these forces point to structually higher Treasury yields.

**European Union:** Faces simulaneous pressure from aging populations, climate-transition Cap-Ex, defense rearmament, and reshoring initiatives. Post-Ukraine defense rearmament adds a significant fiscal burden, and fiscal fragmentation across member states complicates policy responses all combine to expose the EU to broad-based structural inflation pressures.

**United Kingdom:** Faces structural inflation pressures from Brexit-related deglobalization, labor shortages, and high energy transition costs. Post-Brexit labor market tightness is more severe than in continental Europe, while limited fiscal space constrains policy flexibility. High dependence on imported energy and supply chain further increase exposure.

**Japan:** Japan is the most advanced example of aging demographics and fiscal dominance, with debt exceeding 250% of GDP. Although it has historically suppressed yields through monetary policy, the exit from yield curve control marks a major regime shift, with rising inflation and bond yields creating new structural challenges.

**China:** China faces intense pressure from decoupling, deglobalization, and demographic decline, but benefits from stronger state control over finance and industrial policy. While near-term growth challenges remain deflationary, long-term structural forces point toward higher fiscal spending and rising sovereign financing costs.

**Emerging Markets:** Most emerging markets are less exposed to aging populations, climate-transition costs, and reshoring burdens, and some benefit from supply-chain diversification. However, they remain vulnerable to higher global interest rates and reduced capital flows as developed economies absorb more savings to finance their own debt.

# China's Debt-Driven Growth Model

***THE PROPERTY CRISIS AND WEAKENING GOVERNMENT FINANCES ARE RESHAPING CHINA'S ECONOMIC MODEL.***

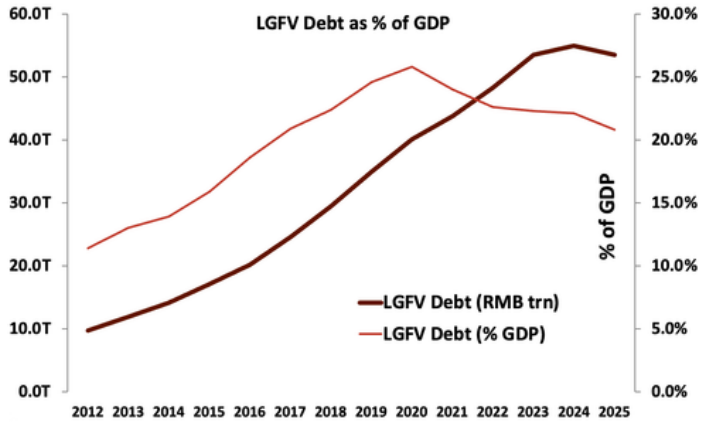
## *THE INFRASTRUCTURE STIMULUS PARADIGM*

In November 2008, the State Council announced a **RMB 4 trillion stimulus** package (13% of 2008 GDP) in response to the global financial crisis. **New yuan loans in 2009 hit RMB 9.6 trillion**, nearly double the prior year. Gross Fixed Capital Formation (GFCF) as a share of GDP rose from **40% to a peak of 48% by 2012**. What was designed as an emergency measure became embedded as the primary growth mechanism.

In 2008, approximately **1.4 units of new credit were needed to generate 1 unit of GDP growth**. By 2020 to 2025, that ratio had widened to an estimated 4 to 5 units of credit per unit of GDP, a fourfold deterioration in capital efficiency reflecting misallocation into low return infrastructure, oversupply of real estate, and zombie like LGFV entities. Debt accumulates faster, but each additional unit of debt generates diminishing economic returns.

## *LOCAL GOVERNMENT FINANCING VEHICLES*

China's 1994 Budget Law prohibited local governments from directly borrowing or running deficits. Yet the 2008 stimulus required local authorities to execute the bulk of infrastructure investment. The solution was the **LGFV, a nominally corporate entity that could borrow on behalf of local government, using state owned land as collateral**. This created a shadow sovereign debt market that operated outside formal fiscal accounting for over a decade.



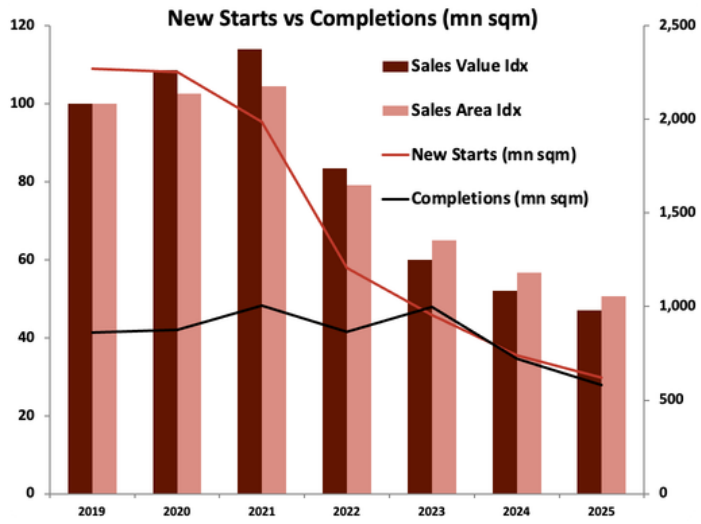

China's property sector, which once accounted for **25–29% of GDP**, was a cornerstone of the country's economy, **supporting local government revenues through land sales, household wealth (with around 70% of household net worth tied to residential property), bank collateral, and the financing of Local Government Financing Vehicles** (LGFVs). As a result, the property downturn became a systemic economic crisis, not just a sector specific slowdown.

The crisis was triggered by the **"Three Red Lines"** policy, which imposed **strict leverage limits on developers**. Since major developers such as Evergrande, Country Garden, Sunac, and Kaisa relied heavily on continuous borrowing, bond refinancing, and presale deposits, the policy simultaneously cut off refinancing across the industry, creating a widespread liquidity crunch. The situation worsened when Country Garden, previously considered one of the healthier developers, defaulted on its USD bond payments in 2023, highlighting that the crisis stemmed from structural weaknesses rather than poor management at a few firms.

# When Debt Stopped Driving Growth

***RISING LEVERAGE HAVE TRANSFORMED CHINA'S GROWTH EQUATION.***

The downturn is expected to be long lasting due to three structural factors: declining demographics, as China's prime homebuying population shrinks and urbanization nears saturation, excess housing inventory, particularly in lower-tier cities where unsold homes represent several years of supply; and permanent balance sheet damage, with developers' equity eroded, the presale financing model losing credibility, and homebuyer confidence remaining weak. Consequently, analysts estimate that China's residential property market will stabilize at **RMB 7 to 9 trillion in annual sales, roughly half of its 2021** peak, indicating that government policies can support stability but are unlikely to restore the sector to its former scale.



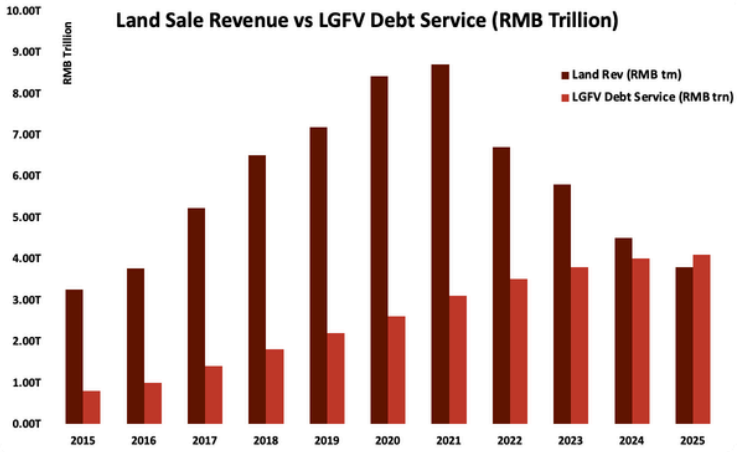

Under China's fiscal system, local governments retain all proceeds from land use rights auctions, making land sales their single largest discretionary income source. Unlike tax revenue, which is shared with the central government on a formulaic basis, land sale proceeds sit entirely with the local authority and can be used flexibly. This made land monetisation the linchpin of local fiscal capacity and the linchpin of the LGFV collateral cycle. It also created a fundamental conflict of interest:

Land sale revenue peaked at **RMB 8.70 trillion in 2021, representing approximately 40% of all local government revenue. By 2025,** it is estimated to have fallen to approximately **RMB 3.80 trillion, a 56% decline in four years,** and now represents only around **19% of local government revenue.** This is not a temporary cyclical dip: it reflects the permanent structural downward shift in property transaction volumes described in The Property Crisis section. The revenue that local governments used to service LGFV debt, fund public services, and maintain infrastructure has been structurally impaired.

As land revenue fell, LGFV debt service obligations did not. In 2021, land revenue of RMB 8.70trn covered LGFV debt service of RMB 3.1trn by a comfortable **2.8× margin**. By 2025, estimated land revenue of around RMB 3.80 trn falls BELOW estimated **LGFV debt service of RMB 4.1 trn, a coverage ratio of approximately 0.9×.** This is the first time in the modern LGFV era that the primary servicing revenue stream has been insufficient to cover the obligations it was designed to support. The gap must be filled by central transfers, additional bond issuance, or, in extremis, restructuring.

# When Debt Stopped Driving Growth

***RISING LEVERAGE HAVE TRANSFORMED CHINA'S GROWTH EQUATION.***



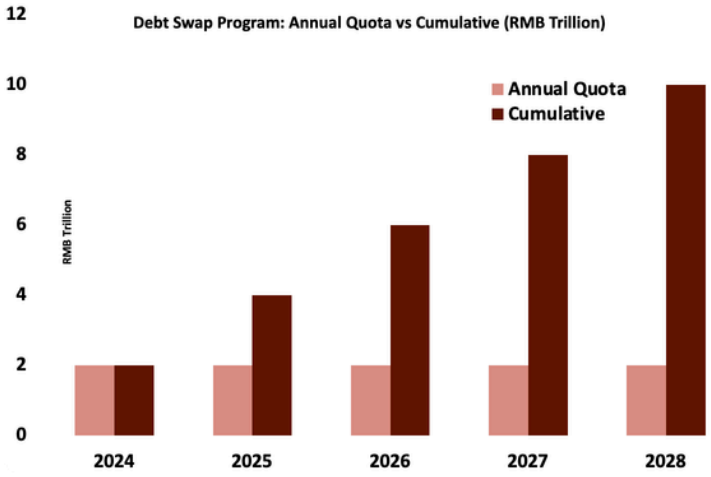

November 2024, the Ministry of Finance announced a RMB 10 trillion debt swap program, the largest single intervention in the LGFV space since a smaller 2015 round. The mechanism: local governments issue formal special purpose bonds **(Local Government Bonds, LGBs) at market rates (2.5 to 3.5%), use proceeds to repay high cost LGFV hidden debt (6 to 7% average)**, and thereby reduce both the interest burden and the refinancing frequency. The program runs at **RMB 2 trillion per annum over 2024–2028. PBoC Governor Pan Gongsheng confirmed in October 2025 that the first tranche had achieved its RMB 2 trillion target.**

The interest rate differential of approximately 350–450 basis points on RMB 10 trillion translates to an annual interest saving of RMB 300–400 billion at full program completion, real and meaningful fiscal relief. Equally important is maturity extension: LGFV hidden debt typically carries 1 to 3 year maturities requiring frequent rollover;

LGBs carry 5–15 year maturities, dramatically reducing the annual refinancing volume and therefore the risk of a rollover driven liquidity crisis.

The swap is a liability management exercise; it changes the cost and tenor of existing debt, not the underlying revenue reality. If land revenues remain at RMB 3.8 to 4.5 trillion annually (the plausible medium term range given structural property dynamics), the coverage deficit remains even after the swap. More critically, 30 to **35% of LGFVs** estimated to have an ICR below **1.0× face a solvency problem**, not merely a liquidity one.

Even reducing interest rates by **350 bps does not restore positive interest coverage** for entities whose revenues have been structurally impaired. These entities require either asset disposals, equity injection, consolidation, or managed wind down, none of which the swap program addresses.



# China Debt Swaps and the Deflation Trap

***BEIJING'S RESPONSE RESTRUCTURES THE DEBT - IT DOES NOT REDUCE IT***

In November 2024, China's NPC Standing Committee approved a RMB 10 trillion ($1.4tn) debt-swap program, converting opaque "hidden" LGFV debt into transparent, lower-cost government bonds. At the same time, a multi-year deflationary stretch is raising the real value of whatever debt remains outstanding - a combination that defines China's current policy bind.

## *EXTEND AND PRETEND*

The swap structure: **RMB 6tn** raises the local debt ceiling over three years, and **RMB 4tn** comes from special-bond allocations through 2028. By the end of 2024, institutions had restructured **RMB 5.4 trillion** of LGFV debt, cutting average rates by **0.76 points and saving roughly RMB 200bn a year in interest.**

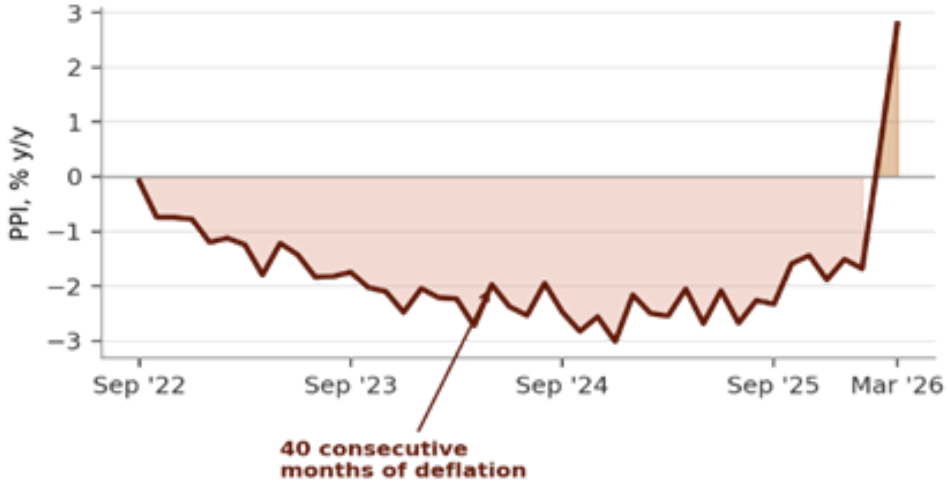
Independent analysts remain skeptical of how much this resolves: Fitch estimates the restructuring covers only about **25% of the true hidden-debt** load, and commentary widely describes the program as "extend and pretend" - lengthening maturities without reducing principal or fixing the revenue gap that created the debt.

## *THE DEBT-DEFLATION TRAP*

China's producer price index fell for 40 consecutive months through December 2025 - the longest streak on record - and the GDP deflator stayed negative for **10 straight quarters**. PPI only turned positive again in March 2026, on a commodity-price rebound whose durability remains uncertain.

Deflation mechanically raises the real value of fixed debt even as the revenue used to service it stagnates. This is the textbook **"debt-deflation trap," and it complicates every deleveraging plan Beijing puts forward.**

**China producer price index (PPI), year-on-year change, illustrative trajectory Sept 2022 - Mar 2026**




Liquidity relief and solvency repair are not the same thing. China has the balance-sheet tools to keep restructuring this debt for years - what it has not yet done is generate the nominal growth needed to outrun it.

# China's Banking and the Backstop

***TWO SLOW-MOVING CONSTRAINTS, AND THE FISCAL CAPACITY THAT BUYS TIME AGAINST BOTH***

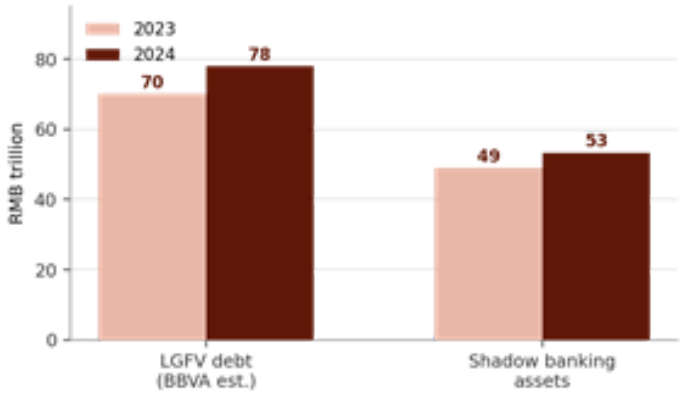
China's banks are more exposed to financing vehicles than to housing directly, and the workforce needed to grow out of all of it has been shrinking since 2011. Both constraints are structural rather than cyclical - and both are, for now, absorbed by a sovereign balance sheet large enough to keep extending timelines.

## *BANKING SYSTEM EXPOSURE*

Banks' LGFV exposure exceeds their direct housing exposure: BBVA Research estimated LGFV debt at **RMB 78 trillion (58% of GDP) as of Q3 2024.**

A parallel shadow-banking system of roughly RMB 53 trillion channels further credit to both LGFVs and developers.

The IMF judged that even a **5% LGFV default rate would lift system-wide non-performing loans by roughly 75%** - a reminder of how concentrated the risk remains despite forbearance holding it in check for now.

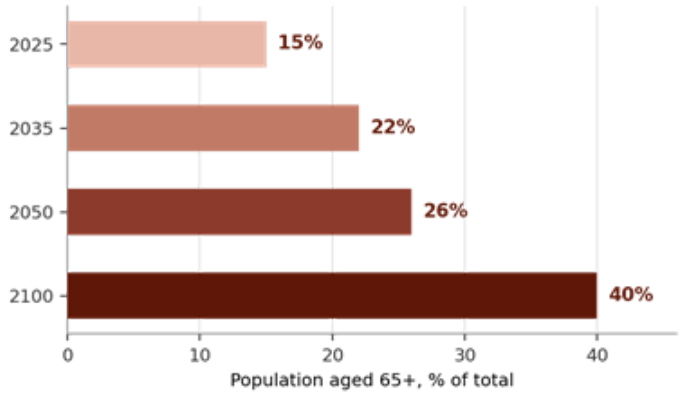



*LGFV debt and shadow-banking assets, RMB trillion*

## *DEMOGRAPHICS: THE SLOW CONSTRAINT*

China's population peaked near **1.4 billion** in 2022 and has declined every year since - the first sustained decline since **1959-1961.**

**The working-age cohort (15-59) had** already begun shrinking back in 2011, with negligible net migration to offset it.




*Population aged 65+, share of total*

Each local-debt or property flare-up is met with central-government swaps, recapitalisation, or special bonds - shifting risk up the sovereign chain rather than resolving it at the source.

The official 2025 budget deficit targeted **~4% of GDP,** but the **IMF's broader augmented** public-sector deficit estimate reaches **roughly 14% of GDP.** A record **$3.4 trillion in FX reserves remains the key external buffer underwriting this capacity.**

# China's Debt Future

***THREE PATHS TO 2035 AND WHERE THE RISK LANDS ACROSS ASSET CLASSES***

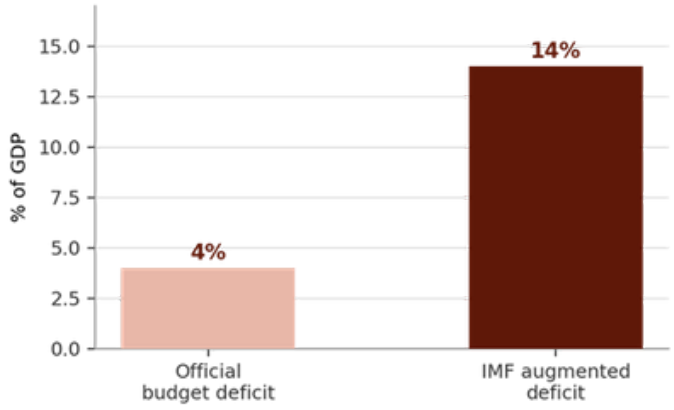
Beijing has the balance-sheet capacity to keep extending timelines for years. The open question for the next decade is whether that capacity buys time for genuine reform, or simply postpones recognition of losses already embedded in the system.

## *SCENARIO ANALYSIS: 2026–2035*

Base Case: Managed deleveraging via repeated debt swaps; growth slows toward the IMF's projected **~3.5% by 2030** as deflation gradually fades.

Bull Case: A record **RMB 160tn in household savings funds** a consumption-led rebalancing; productivity gains partly offset demographics; LGFV debt stabilises near current levels.

Bear Case: A Vanke-scale default recurs at greater scale or spreads to multiple regional banks, forcing direct recapitalisation and pushing augmented debt toward **140-150% of GDP.**




***2025 fiscal deficit: official vs. IMF augmented, % of GDP***

## *INVESTMENT IMPLICATIONS*

Equities: Domestic-consumption and tech/EV names vs. property-linked developers, building materials and regional bank exposure.

Fixed Income: Credit differentiation matters: central govt and coastal LGFV paper carries implicit support; inland/poorer-province LGFV debt does not.

Commodities: Construction-linked demand (steel, cement) structurally lower as residential starts fall **~20%/yr; export-led manufacturing provides a partial offset.**

Currencies & Gold: RMB held broadly stable by PBoC management; gold has been a clear beneficiary of household de-risking, with jewellery prices up **~68% y/y in Dec 2025.**

China does not face a conventional sovereign debt crisis. Instead, it faces the challenge of managing one of the world's largest hidden leverage structures while transitioning to a fundamentally different economic model. The success or failure of this transition will likely determine China's growth trajectory for the next decade.

# US : Rising Rates & Structural Pressures

***THREE PATHS TO 2035 AND WHERE THE RISK LANDS ACROSS ASSET CLASSES***

## *THE END OF THE ZERO-RATE FISCAL REGIME*

For the better part of a decade following the Global Financial Crisis of 2008–2009, the United States financed its expanding public debt in an historically anomalous environment: real interest rates near or below zero, and nominal rates suppressed by the Federal Reserve's unprecedented monetary accommodation.

The federal funds rate was pinned at the effective lower bound **(0–0.25%) from December 2008** through December 2015, and again from March 2020 through March 2022.

Inflation-indexed 10-year Treasury yields averaged near zero from 2012 to 2021, having fallen from over **2% during 2003–07.**

This environment fundamentally distorted fiscal policymaking calculus. With real rates below nominal **GDP growth (r < g), deficit financing appeared self-liquidating.** The result was a decade-long expansion in federal borrowing with minimal market penalty.

The inflation surge of 2021–2022 decisively broke the zero-rate regime. The Federal Reserve raised the federal funds rate from near zero in March 2022 to **a 23-year peak of 5.25–5.50% by July 2023, where it remained until September 2024.**

The Fed then cut rates by a cumulative **100 basis points through December 2024 (to 4.25–4.50%) and held the target at 3.50–3.75% into 2026.**

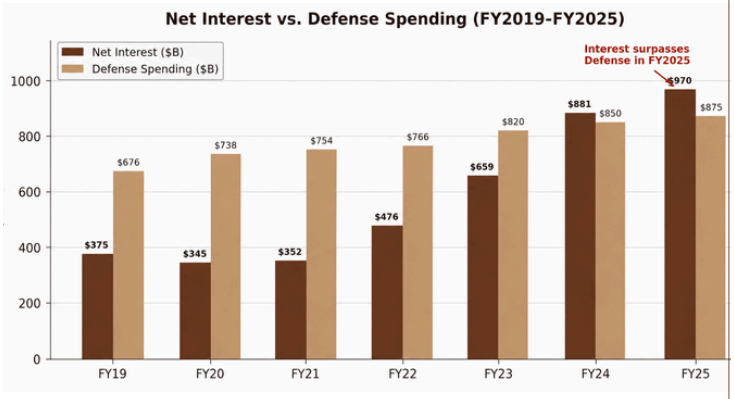
As pandemic-era borrowing rolled over into the 4–5% rate environment, annual interest outlays more than doubled in three years: **from $476 billion in FY2022 to $970 billion in FY2025**.

Each **1% increase in the average interest rate on $39 trillion in federal debt adds approximately $390 billion per year** in additional annual interest expense, permanently compounding future deficits. ***(Source: US Fiscal Clock, 2026)***

## *INTEREST COSTS AS STRUCTURAL BUDGET DRIVER*

For the first time in American history, net interest payments on the public debt surpassed $1 trillion in FY2025. The CBO Monthly Budget Review confirmed that gross interest payments on the public debt exceeded **$1 trillion, with the net figure reaching $970 billion, a 10% year-over-year increase from $881 billion in FY2024.**

Five years earlier (FY2020), net interest was only $345 billion. The **2.8x multiplication in five years** was driven almost entirely by the interaction of a larger debt stock and substantially higher rates.



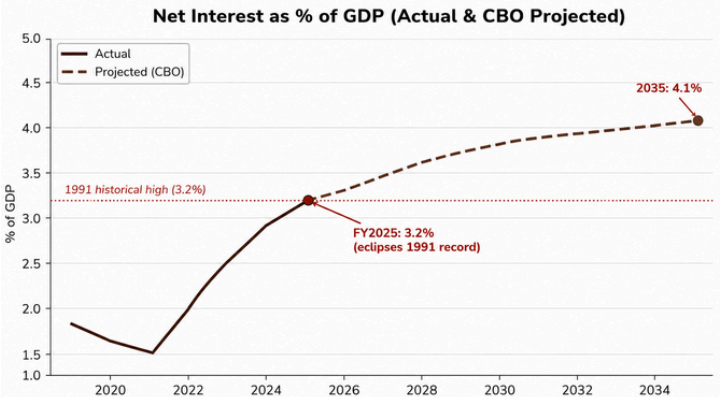

Each local-debt or property flare-up is met with central-government swaps, recapitalisation, or special bonds - shifting risk up the sovereign chain rather than resolving it at the source. The official 2025 budget deficit targeted ~4% of GDP, but the IMF's broader augmented public-sector deficit estimate reaches roughly 14% of GDP. A record $3.4 trillion in FX reserves remains the key external buffer underwriting this capacity.

# US : Rising Rates & Structural Pressures

***THREE PATHS TO 2035 AND WHERE THE RISK LANDS ACROSS ASSET CLASSES***

CBO projects net interest will be the fastest-growing major spending category through 2035, rising 76% from **~$1.0T (FY2026) to $1.8T (FY2035).** Social Security grows **58% and Medicare 75% over the same period.**

The **NBER (2026)** estimates net interest will account for more than **100% of the increase in the unified deficit through 2054**, meaning a primary balance alone would not stabilize the debt trajectory.




**Source: CBO Budget & Economic Outlook 2025–2035; CBO Long-Term Budget Outlook 2025–2055; PGPF (2026)**

Budget flexibility- **$7,300 per household**, the FY2025 share of net interest payments.

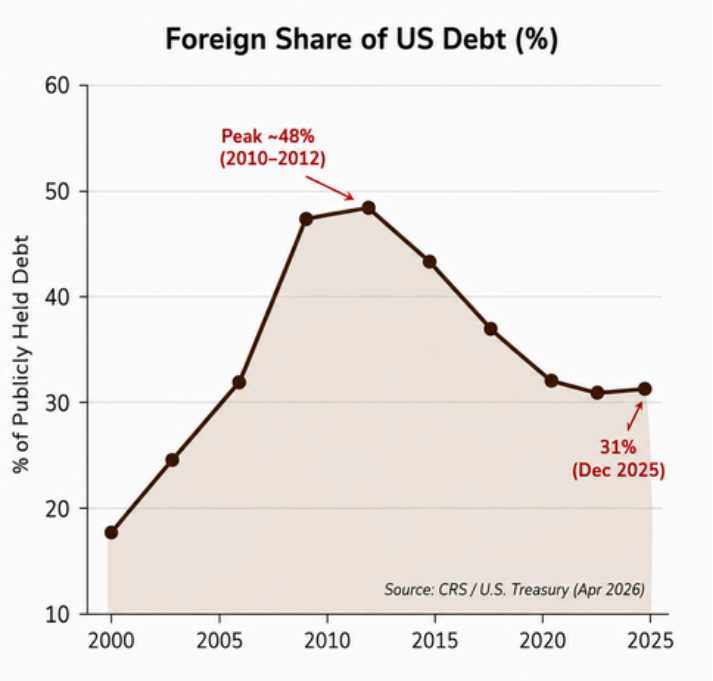
More than average household spending on healthcare ($6,500), home furnishings ($2,600), gasoline ($2,500), clothing ($2,200), and education ($1,200) combined. ***(Source: American Action Forum, 2025)***

The U.S. government must simultaneously refinance maturing debt and fund new deficits, creating gross issuance requirements that reached unprecedented levels in 2025.

Approximately **$9.2 trillion in marketable Treasury securities** matured in FY2025, roughly one-third of all outstanding debt and nearly 30% of GDP, with an estimated **55–60% falling due before July 2025.**

## *TREASURY SUPPLY & MARKET ABSORB CAPACITY*

Adding the **$1.9 trillion FY2025** deficit placed gross issuance above $10 trillion: a volume no modern **capital market had previously absorbed in a single fiscal year.**



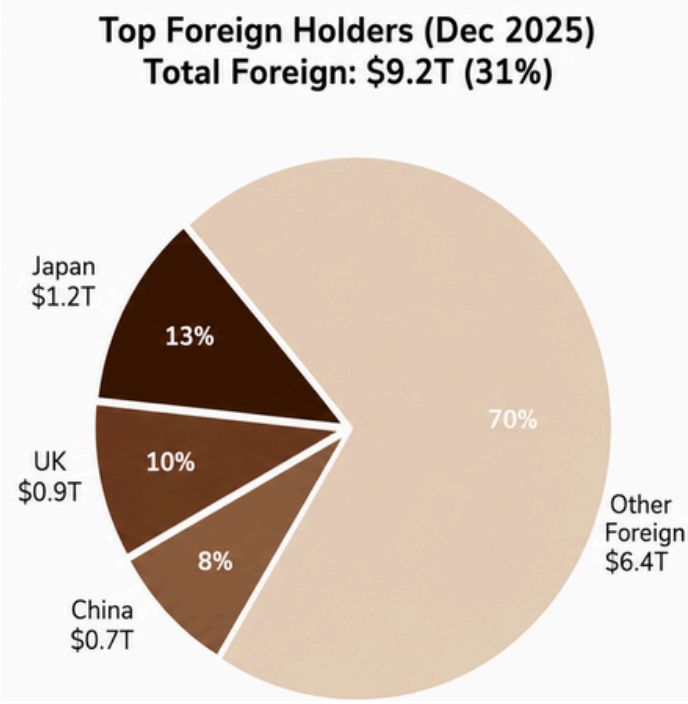

**Source: Congressional Research Service (Apr 2026); U.S. Treasury TIC Data;**

Research from the Kansas City Federal Reserve (2026) provides the most rigorous recent estimate of yield sensitivity to foreign selling: a one standard deviation increase in monthly foreign Treasury liquidation **(~$141 billion) is estimated to raise 10-year yields by an average of 57 basis points,** with estimates ranging from 25 to 101 basis points.

With the Fed running down its balance sheet and foreign official demand structurally constrained, domestic private investors must absorb an increasingly **large share of Treasury supply at a price.**

# US : Rising Rates & Structural Pressures

***THREE PATHS TO 2035 AND WHERE THE RISK LANDS ACROSS ASSET CLASSES***




**Source: Congressional Research Service (Apr 2026); U.S. Treasury TIC Data;**

Market risk- The U.S. government must simultaneously refinance maturing debt and fund new deficits, creating gross issuance requirements that reached unprecedented levels in 2025.

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## *THE MATURITY CHALLENGE:*

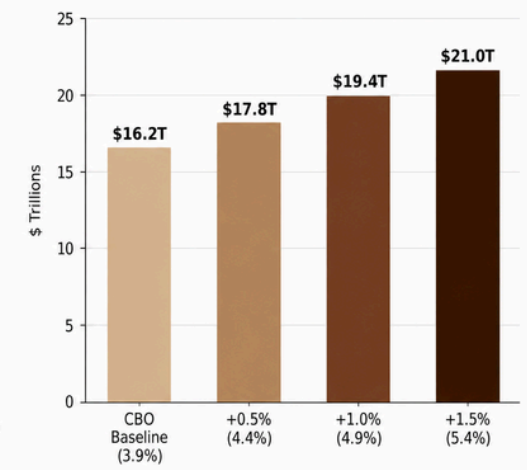
The maturity structure of U.S. Treasury debt determines how quickly changes in market rates translate into changes in federal borrowing costs. **As of mid-2025, over 80% of outstanding federal debt will mature within 10 years, and approximately 61% matures by end-FY2028.**

Over **20% required refinancing** in FY2025 alone. The **weighted average maturity** sits near **72 months, close to a three-decade high** , owing to deliberate **'terming out' executed during 2020–2021 when long-term rates were near historical lows.**

The interest cost trajectory is highly sensitive to rate assumptions. Under CBO's January 2025 baseline, the average annual rate on **10-year Treasury notes was projected at 3.9% for 2025–2035.**

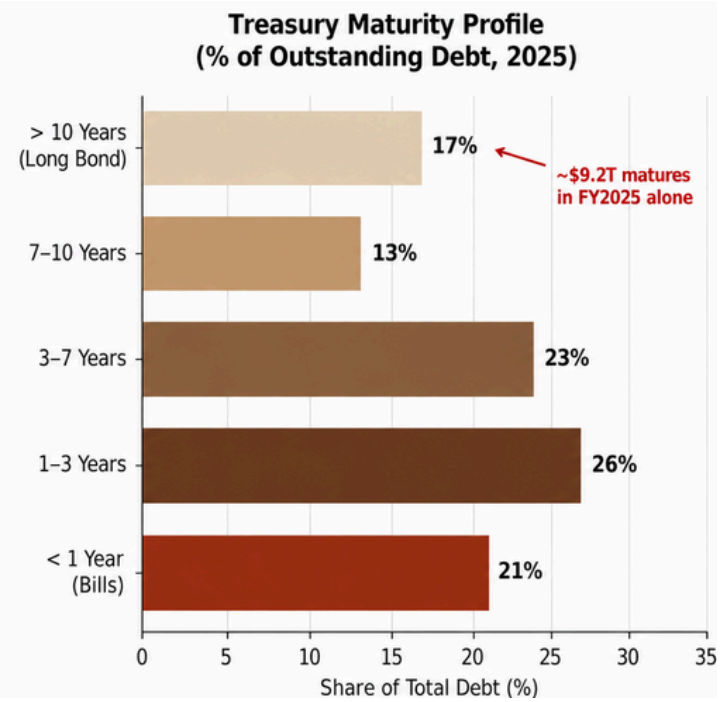
With the 10-year yield trading above **4.5% as of mid-2026,** EPIC for America estimates that a 1% persistent upward shift adds **$3.2 trillion in cumulative interest costs over the decade.**

A critical structural insight: funding costs may continue rising even if policy rates stabilize, as older low-coupon debt (**issued at sub-1% yields in 2020–2021) matures and is replaced by current-market instruments yielding 4–5%.**



# US : Rising Rates & Structural Pressures

***THREE PATHS TO 2035 AND WHERE THE RISK LANDS ACROSS ASSET CLASSES***




**Source: U.S. Treasury Office of Debt Management; Econofact (Jul 2025); EPIC for America (2026); CBO (Jan 2025 baseline)**

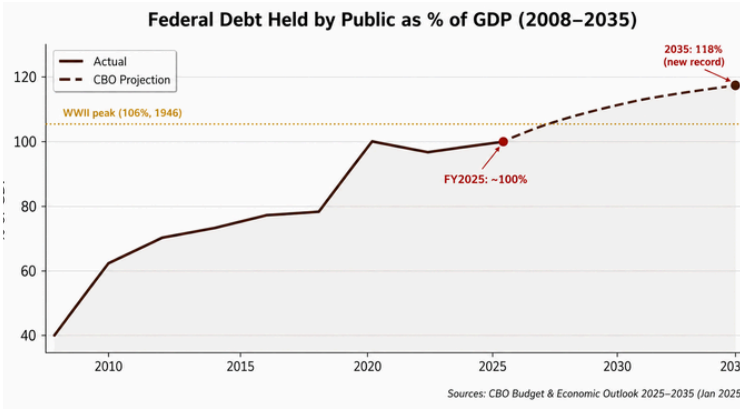
A sustained **1 percentage point upward** deviation from CBO's rate baseline adds **$3.2 trillion** in cumulative federal **interest costs over 2025–2035,** equivalent to **15%+** of the entire baseline deficit projection. (Source: EPIC for America, 2026)

## *THE $100T FISCAL ICEBERG*

Fiscal dominance is a macroeconomic regime in which the size and financing requirements of government debt constrain the central bank's ability to conduct independent monetary policy.

In its classical formulation **(Sargent & Wallace, 1981), fiscal dominance occurs when raising rates to control inflation.**

Forcing the central bank to eventually monetize debt or accept **higher inflation to reduce the real debt burden.**




TAt a **$39+ trillion debt stock**, every 100 basis points of additional rate tightening adds approximately $390 billion in annual interest expense, creating an implicit constraint on the Fed's willingness to sustain restrictive policy.

When the Fed raised rates to **5.25–5.50% in 2023, the fiscal cost manifested as a jump in net interest from $476 billion (FY2022) to $970 billion (FY2025).** Further tightening would impose even larger fiscal costs

## *THE ICEBERG BENEATH THE DEBT NUMBER*

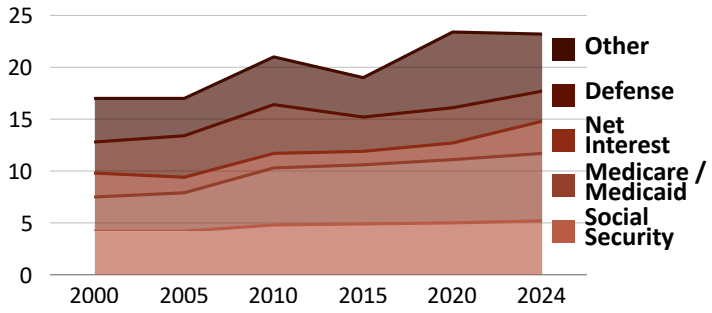
The **$36 trillion** headline national debt is the visible portion of America's fiscal exposure. Present-value estimates of unfunded Social Security and Medicare obligations **add a further $50–100+ trillion in commitments that appear in no deficit headline yet fall due with actuarial certainty as the population ages.**

In FY 2025, mandatory outlays totalled **$4.2 trillion, of which Social Security ($1.5T) and Medicare/Medicaid ($1.7T) together consumed 76 cents of every mandatory dollar.**

# US : Rising Rates & Structural Pressures

***THREE PATHS TO 2035 AND WHERE THE RISK LANDS ACROSS ASSET CLASSES***

*Evolution of U.S. Federal Spending by Category (% of GDP)*




Under CBO's 2025–2055 baseline, Medicare and Medicaid spending climbs from roughly **5.8% of GDP today toward 8.1%** by 2055, equivalent, as a share of the economy, to the entire federal budget in 2007.

Net interest simultaneously reaches **5.4% of GDP by 2055, exceeding Medicare spending that year.**

**Social Security, Medicare, and net interest alone grow from 49% of all federal expenditures in 2025 to 58% by 2035,** crowding out defence, infrastructure, and every other discretionary priority.

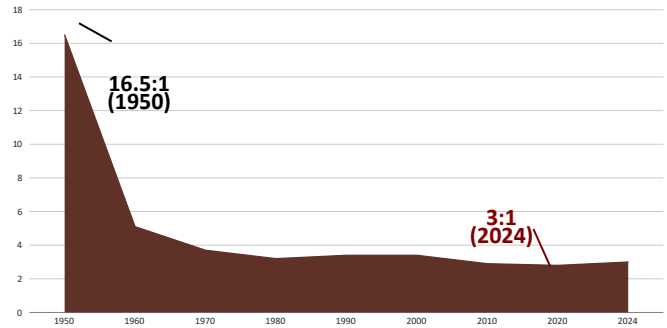
## *THE DEMOGRAPHIC ENGINE*

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The root cause is structural, not cyclical. In 1960, 5.1 workers funded each Social Security beneficiary sufficient to sustain the programe comfortably.

*U.S. Worker-to-Beneficiary Ratio (1950-2024)*




**By 2024 that ratio stood at 3.0:1,** already below the pay-as-you-go break-even of 2.8. SSA Trustees project it falls to **2.5:1 by mid-century.**

Peak 65' (2024-2027) sees **4.1 million Americans turning 65 annually the largest retirement surge in U.S. history.**

The combined trust fund is projected to exhaust in 2033, triggering automatic **23% benefit cuts absent congressional action. This is arithmetic, not policy failure.**

## *RISK BEFORE IT APPEARS IN CREDIT RATINGS*

Credit rating downgrades tend to lag market signals. Although Moody's downgraded the U.S. from Aaa to Aa1 in May 2025, financial markets had already begun pricing higher sovereign risk. Three indicators reflected this shift:

net interest payments reached **$970 billion in FY2025 (18.5% of federal revenue) and are projected to rise to $1.8 trillion by 2035.**

# Dollar Dominance and the Decade Ahead

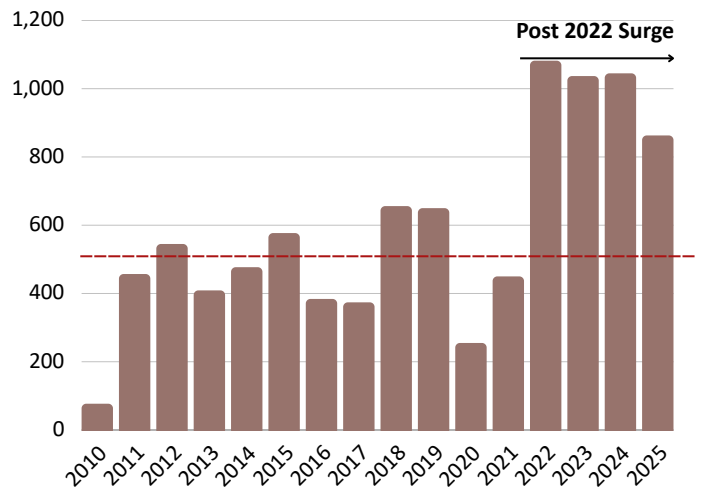
***ASSESSING THE DURABILITY OF RESERVE CURRENCY PRIVILEGE***

## *GLOBAL CAPITAL FLOWS & DOLLAR*

The dollar's reserve status grants the U.S. what economists call an "exorbitant privilege" the ability to borrow cheaply in its own currency with virtually unlimited global demand absorbing its deficits.

That privilege is showing signs of structural erosion. The dollar's share of global foreign-exchange reserves has declined from **71% in 2000 to 58% today (IMF COFER data)** , slow in absolute terms but consistent and unidirectional across every single year of measurement

*Global Central Bank Gold Purchases (2010–2025)*

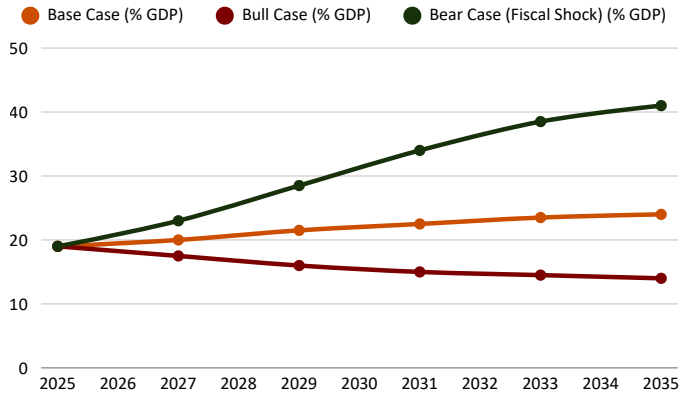



The 2022 inflection. When the U.S. and allies froze **$300 billion of Russia's sovereign dollar reserves following the Ukraine invasion, central banks globally absorbed a visceral lesson:** dollar reserves held abroad carry political seizure risk. Gold does not, it has no counterparty, cannot be frozen, and cannot be printed.

The result: three consecutive years (2022-2024) of central bank gold purchases exceeding **1,000 tonnes annually, against an average of just 473 tonnes from 2010-2021.**

China, India, Poland, and Turkey led accumulation. India repatriated 100 tonnes from Bank of England vaults in 2024. A World Gold Council survey (2024) finds 76% of central banks expect gold to form a higher share of reserves within five years; **73% expect the dollar's share to shrink**, deliberate, institution-level portfolio rebalancing, not panic.

## *SCENARIO ANALYSIS: THE NEXT DECADE (2025-35)*




Base Case: Controlled Deterioration (55%) Deficits persist at **5-6% of GDP.**

Debt reaches 120% of GDP by 2036 (CBO Feb 2026 baseline, incorporating the One Big Beautiful Bill Act which adds **$4.2 trillion to deficits through 2034). Interest absorbs 22-24% of revenue by 2035. Growth holds near 1.8-2%.**

No acute crisis, but policy optionality erodes steadily less room to absorb the next recession or geopolitical shock each passing year.

Bull Case: **AI Productivity Escape (25%)** Broad-based AI-driven productivity adds 0.5-1pp to annual real GDP growth.

Higher nominal GDP lifts tax receipts without rate increases, stabilising **debt/GDP below 110%** and allowing interest costs to recede as a share of revenue.

# Market Of Sovereign Stress

***ASSESSING THE DURABILITY OF RESERVE CURRENCY PRIVILEGE***

Historical precedent: the 1990s tech productivity boom turned structural deficits into surpluses by decade's end. The critical variable is whether AI gains show up in tax receipts not merely in equity valuations.

Bear Case: Fiscal-Financing Shock **(20%) Persistent** deficits trigger a meaningful yield spike as Treasury demand softens. **With $9+ trillion maturing within three years, rollover at higher rates rapidly accelerates debt-service costs.**

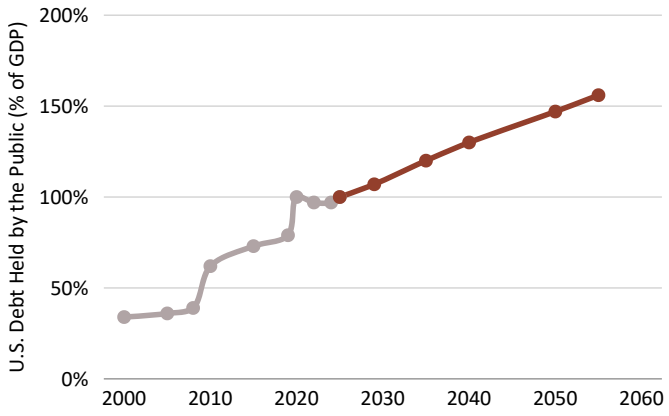
The risk is a self-reinforcing loop: higher deficits fi more issuance fi higher yields demanded fi still-higher deficits.

The U.S. would not formally default (it prints dollars), but monetisation or forced austerity carries its own severe economic costs.

Reserve diversification does not threaten dollar primacy overnight; no credible alternative exists at scale.

But each marginal shift in allocation raises the clearing yield the **U.S. must offer on new debt, compounding into meaningfully higher funding costs over a decade of persistent borrowing.**

## *DEBT TRAJECTORY: CBO BASELINE PROJECTION*




***U.S. Debt Held by the Public: Historical Trend and Long-Term Projections (2000-2055)***

Federal debt held by the public reached 100% of GDP in 2025, surpasses the **post-WWII record of 106% in 2029, and hits 156% by 2055 under the March 2025 baseline.**

The February **2026 CBO update revises** this to **175% of GDP by 2056. The deficit averages 6.3% of GDP over 30 years, more than 1.5× the 50-year historical average of 3.8%.**

The sovereign debt challenge did not emerge when governments accumulated debt, it emerged when higher rates ended the era of cheap money and refinancing costs began compounding. **Fiscal space is now structural, not cyclical.**

## *POSITIONING FOR THE NEW REGIME*

***RATES / FIXED INCOME : Higher-for-Longer Risk***

Long-duration Treasuries carry asymmetric risk in this regime. The bull-case upside requires an AI productivity miracle, while base and bear cases both produce persistent term premium expansion.

With the **Fed holding at 3.75%** and PCE inflation revised to **3.6% for 2026,** the path to meaningfully lower rates is neither clear nor near-term.

Positioning: Underweight long duration. Favour short-dated T-bills and floating-rate exposure. TIPS for inflation protection.

***CREDIT : Sovereign Crowding-Out***

When the **U.S. government borrows trillions annually** it competes with the private sector for the same capital pool.

# Market Of Sovereign Stress

***ASSESSING THE DURABILITY OF RESERVE CURRENCY PRIVILEGE***

***REAL ASSETS : Inflation-Sensitive Positioning***

Persistent deficits carry latent inflationary risk: physical assets that rise with price levels provide a natural hedge. The bear-case fiscal shock scenario - monetisation as a policy tool produces meaningfully higher realised inflation than consensus projects.

**Positioning:** Commodities, infrastructure (toll roads, energy grid), farmland, inflation-linked bonds (TIPS).

***GOLD : Strategic Hedge Against Fiscal Uncertainty***

Gold has no counterparty, cannot be frozen, and cannot be printed, exactly the properties central banks discovered they needed in 2022.

It provides cross-scenario protection against dollar confidence erosion, fiscal dominance, and latent inflation simultaneously. **Three consecutive years of record central bank purchases signal institutional conviction.**

**Positioning:** Strategic **5–10% allocation. Portfolio insurance, not a directional bet.**

***EQUITIES : Fiscal Beneficiaries vs. Rate-Sensitives***

Not all equities are equal. Fiscal beneficiaries: defence contractors, infrastructure operators, healthcare providers are structurally supported by the very deficits that stress the broader picture.

Rate-sensitives REITs, utilities, long-duration growth names, heavily indebted corporates, face sustained headwinds from elevated financing costs. Positioning: Overweight defence, infrastructure, healthcare. Underweight utilities, REITs, high-leverage growth

## *THE END OF FISCAL ABUNDANCE*

The United States does not face an imminent solvency crisis.

What it faces is a gradual, compounding transition from an era of **near-free capital** and abundant fiscal flexibility toward one of structurally higher funding costs, shrinking policy optionality, and growing dependence on sustained global investor confidence.

Demographics will not reverse. **Interest obligations are contractual.**

Global capital is slowly but **deliberately diversifying.**

Markets priced the transition well before rating agencies acted.

The defining question of the next decade is not whether **U.S. debt is sustainable**, but under what conditions, at what cost, and with how much remaining room to absorb the crises yet unforeseen.

# Origins of Japan's Debt Accumulation

***HOW STAGNATION TRANSFORMED JAPAN INTO THE WORLD'S MOST INDEBTED MAJOR ECONOMY.***

Japan's debt ratio was below 68% of GDP in 1991, comparable to France or the United Kingdom at the time. Japan's path to 263% debt-to-GDP wasn't a single policy failure, it was a 35-year compounding of four mutually reinforcing forces

## POST-BUBBLE ECONOMIC STAGNATION

Late 80s bubble: The Nikkei 225 tripled, and Tokyo land prices reached extreme valuations.

Bubble burst (1989-90): BoJ hike rates from 2.5% to 6% within a year, triggers **60% Nikkei crash, 87% land price decline, and ¥1,500 Tn wealth loss.**

Aftermath: Balance sheet recession and zombie lending kept real GDP growth below **1%, while nominal GDP fell from ¥521 Tn (1997) to ¥475 Tn (2012), worsening debt.**

## FISCAL STIMULUS PROGRAMS

Fiscal stimulus (1992-99): Japan announces 8 stimulus packages totaling **¥100 Tn, primarily funding public infrastructure projects.**

Policy setbacks: 1997 consumption tax hike (3% to 5%) coincided with the Asian Financial Crisis, pushing GDP down **2.5% (1998) and forcing renewed stimulus.**

Long-term impact: Further spending after the 2001 IT crash and 2008 **Global Financial Crisis raised debt (~¥25 Tn)** but poorly targeted investments produced weak economic returns and limited growth.

| Year | Event | Debt/GDP |
|---|---|---|
| 1991 | Asset bubble collapse, fiscal stimulus begins | ~68% |
| 1997 | Asian Financial Crisis, banking crisis deepens | ~98% |
| 2000 | Dot-com slowdown, additional stimulus packages | ~135% |
| 2008-09 | Global Financial Crisis, emergency spending | ~180% |
| 2011 | Triple Disaster (earthquake, tsunami, Fukushima) | ~210% |
| 2013 | Abenomics launched; debt crosses ¥1 Qn | ~230% |
| 2020 | COVID-19 pandemic emergency expenditure | ~256% |
| 2024 | Debt reaches ¥1.32 Qn | ~263% |

***Debt Accumulation Timeline***

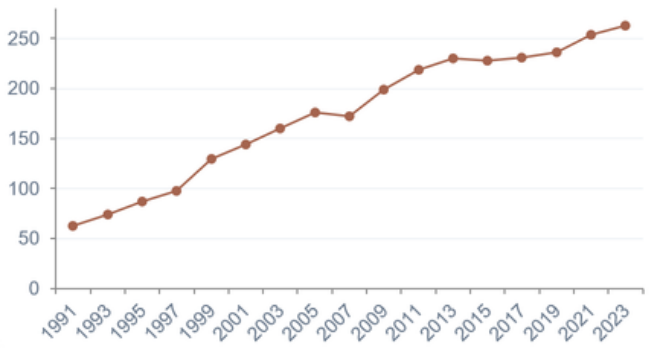
## DEFLATIONARY PRESSURES

Persistent deflation (1994-2012): Core CPI turns negative 1998, **prices falling 0.5-1.5% annually, reducing consumption and investment.**

Debt burden: Deflation increased real value of debt, prolonged zombie lending. Kept nominal GDP stagnant despite positive real GDP growth.

## GOVERNMENT BORROWING EXPANSION

Rapid debt growth: Govt. debt rose from **68% of GDP (1991) to 263% (2024)** due to **persistent primary deficits and continuous borrowing.**



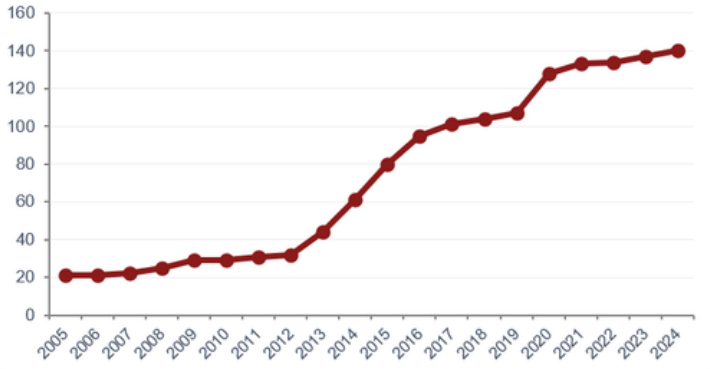

***Japan Gross Debt-to-GDP (%)***

# Origins of Japan's Debt Accumulation

## THE BANK OF JAPAN AND YIELD CURVE CONTROL

Japan's central bank has evolved from lender of last resort into the dominant price-setter across its own sovereign bond market. Through successive rounds of quantitative easing and introduction of Yield Curve Control in 2016, BoJ now owns nearly half of all outstanding JGBs, a concentration of market power without precedent among major economies.

***BOJ BALANCE SHEET (% OF GDP)***




***Quantitative Easing - The Foundation***

- **BoJ launches first QE program in 2001,**
- 2013's QQE dramatically escalates this, committing double monetary base in two years **(¥50Tn/year JGB purchases).**
- Tranforms BoJ from market player to marginal price-setter, removing risk premium.

***KEY INSIGHT***

BoJ has become the dominant participant in its own sovereign bond market, owning half of all JGBs, setting yields by fiat. It is not a participant in the JGB market, it IS the market.

***Yield Curve Control***

- From Sept 2016, BoJ pegged the **10 yr JGB yield at ~0%** becoming an unlimited buyer at ceiling price.
- Tolerance band widened progressively **(+/-0.1→+/-1%)**, but each widening required massive purchases.
- At peak, **BoJ bought 78% of gross JGB issuance in a year.**

## THE TRAPPED EQUILIBRIUM

Japan faces a paradox with no easy exit. Policies that make its debt serviceable today such as the near zero interest rates sustained by BoJ purchases, are incompatible with a functioning bond market, sustainable monetary policy, and long term financial stability. While raising rates risks fiscal crisis, keeping them low risks currency collapse and market dysfunction.

| *Fiscal-Monetary Interdependence* | *Debt Servicing Sensitivity* |
|---|---|
| The Ministry of Finance government bonds are purchased by BoJ to suppress bond yields and keep borrowing costs low. | With gross debt at ¥1.32 Quadrillion, the interest rate sensitivity is acute. At: **~¥26 Tn (~19% of revenues)** **3%: ~¥40 Tn (~28% of revenues)** |

# Origins of Japan's Debt Accumulation

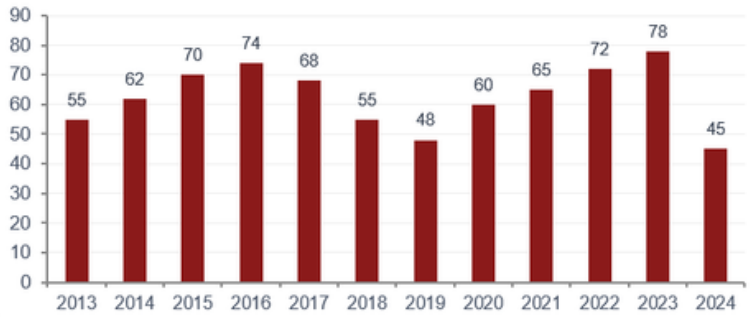
***THE BANK OF JAPAN AND YIELD CURVE CONTROL***

Japan's central bank has evolved from lender of last resort into the dominant price-setter across its own sovereign bond market. Through successive rounds of quantitative easing and introduction of Yield Curve Control in 2016, BoJ now owns nearly half of all outstanding JGBs, a concentration of market power without precedent among major economies.

## *POLICY CONSTRAINT MATRIX*

| Policy Action | Fiscal Impact | Currency Effect | Market Effect | Verdict |
|---|---|---|---|---|
| Raise rates aggressively | Debt service spike | Yen recovers | JGB sell-off | Destabilising |
| Maintain near-zero rates | Costs contained | Yen collapse | No price discovery | Unsustainable |
| Gradual normalization | Manageable if slow | Mixed | Volatile transition | Current path |
| Resume YCC | Costs suppressed | Currency crisis | Market dysfunction | High risk |

## *MARKET FUNCTIONING*




***BoJ % of Gross JGB Insurance Purchased***

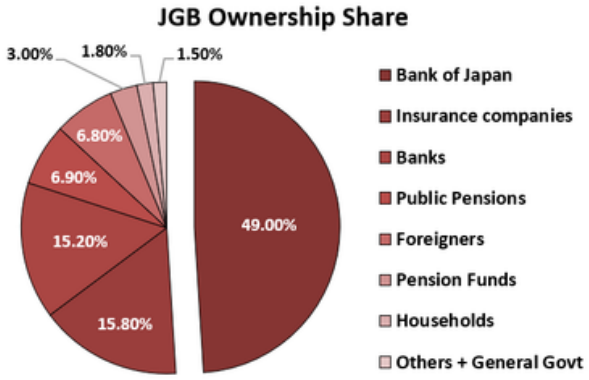
***Liquidity Destruction:***

Bid-ask spreads on 10yr JGBs widened ~400% from 2016 levels. The BoJ's market function index reached −24 in 2023, the most impaired reading ever recorded. **Days with zero JGB trades became common.**

***Taper Challenge:***

As BoJ reduced bond purchases **(¥400Bn to ¥200Bn/month by Jul 2025),** private investors had to absorb larger share of JGB issuance.

Hedge funds have significantly increased leveraged JGB positions through repo market, creating a new source of systemic volatility that the **BoJ explicitly flaggged in Oct 2025.**



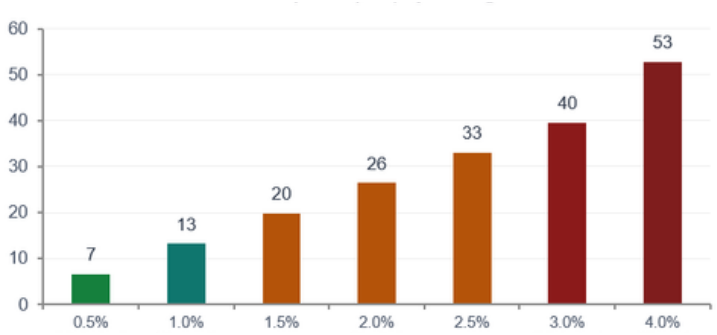

Japan's debt has stayed stable at extreme levels because it is owed almost entirely to itself. Around 93% of JGBs are held domestically and only ~7% by foreigners, insulating Japan from the capital flight that destabilizes most indebted nations.

# The Cost of Higher Interest Rates

***THE BANK OF JAPAN AND YIELD CURVE CONTROL***

With gross government debt at ¥1.32 Quadrillion, Japan's interest rate sensitivity is the defining fiscal vulnerability of its balance sheet. Even modest increases in the average cost of funding translate into fiscally material shifts in interest expenditure, particularly as the stock of low-coupon ZIRP-era bonds is progressively refinanced at higher market rates.

## *INTEREST RATE SENSITIVITY ANALYSIS*




***Annual Interest Expense (¥Tn) by Avg JGB Yield***

- Interest costs increase almost linearly as average JGB yields rise.
- Every 1 percentage point (100 bps) increase in average borrowing costs adds roughly **¥13 trillion** to annual interest expenditure.
- A return to historically normal interest rates **(~3%) would raise annual debt servicing costs to ~¥40 trillion,** placing significant pressure on Japan's fiscal position.
- Every 100 bps (1%) increase in borrowing costs raises interest expenditure by roughly **10 percentage points of government revenue.**

## *REFINANCING MECHANICS*

- Japan's average JGB maturity is 8-10 years, meaning approx **10-15%** of total stock matures annually and must be refinanced at prevailing market rates.
- Annual gross JGB issuance for FY2026 is planned at ¥180.7 Tn, covering both **rollovers and new borrowing.**
- The BoJ's ongoing taper reduces monthly JGB purchases by a cumulative **¥2.4 trillion per year,** requiring private investors to absorb additional government bond supply.
- Debt service costs are projected to rise from **~4.5% of GDP now toward ~9%** by 2035 under normalization scenarios.

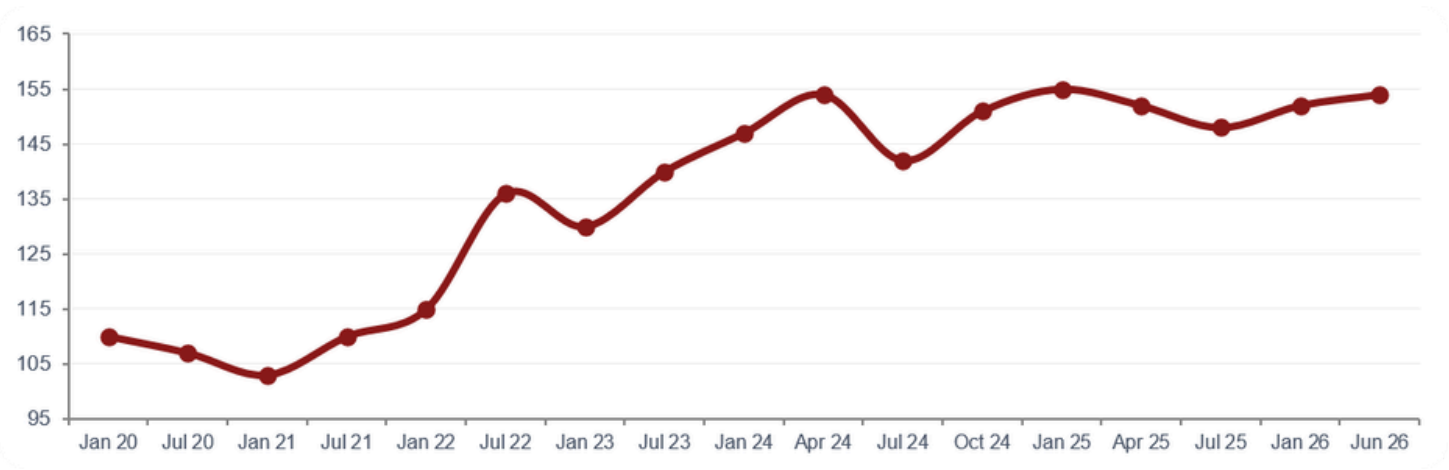
## *BUDGETARY IMPACT AND MITIGANTS*

- Normalization to 2% average yields would add ~¥13Tn net after accounting for the BoJ remittance offset **(the BoJ returns ~50% of interest income to the Treasury).**

Every 100bps rise in average JGB yield adds approximately **¥13.2 trillion to annual debt service**, equivalent to raising Japan's entire defense budget four times over in a single policy move.

# Yen @ The Pressure Valve

***THE BANK OF JAPAN AND YIELD CURVE CONTROL***




***Yen Depreciation Trends (USD/JPY)***

- Import Inflation: Japan's dependence on imported energy and commodities meant a **30%+ yen depreciation since 2020 drove persistent inflation, forcing the BoJ toward policy normalization despite political opposition.**

- Capital Flows & Carry Trade: Years of **near-zero interest rates made the yen the world's primary funding currency,** making the unwinding of carry trades a source of significant global market volatility as Japan normalized policy.

- Exchange-Rate Dynamics: Persistent structural capital outflows have kept the yen weak despite record FX interventions, showing that **exchange-rate pressures** are driven by long-term investment flows as much as interest-rate differentials.

- The Normalization Paradox: Japan faces an unavoidable trade-off: **higher interest rates strengthen the yen but increase debt servicing costs**, while lower rates support fiscal sustainability at the expense of inflation and currency weakness.

| USD/JPY Level | Policy Stance | Political Mood | Likely Response |
|---|---|---|---|
| Below 140 | Comfortable, BoJ can pause | Consumers benefit | No action needed |
| 140 - 155 | Tolerable, monitor closely | Mixed signals | Verbal warnings only |
| 155 - 165 | Uncomfortable, CPI risk | Cost-of-living pressure | FX intervention + rate hike signal |
| Above 165 | Crisis, 1998-style risk | Political emergency | Coordinated intervention + emergency hike |

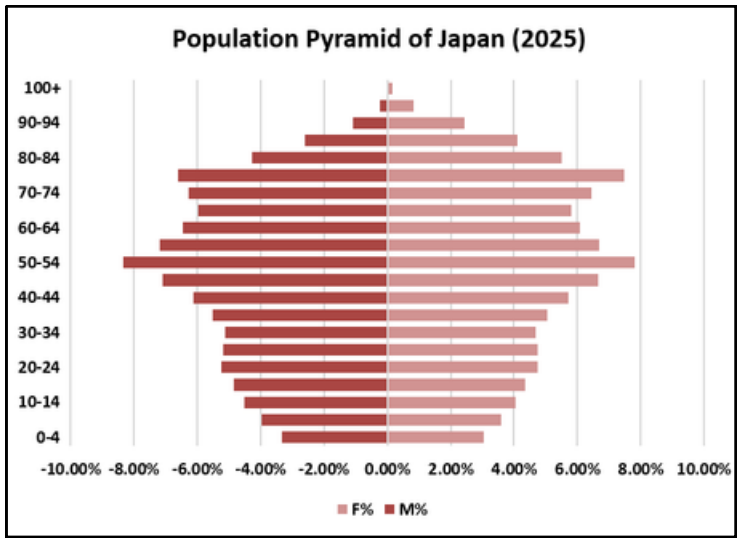
- Key mitigants: nominal **GDP growth of 4% p.a.** provides denominator relief, fixed-rate bonds reprice slowly over a decade.

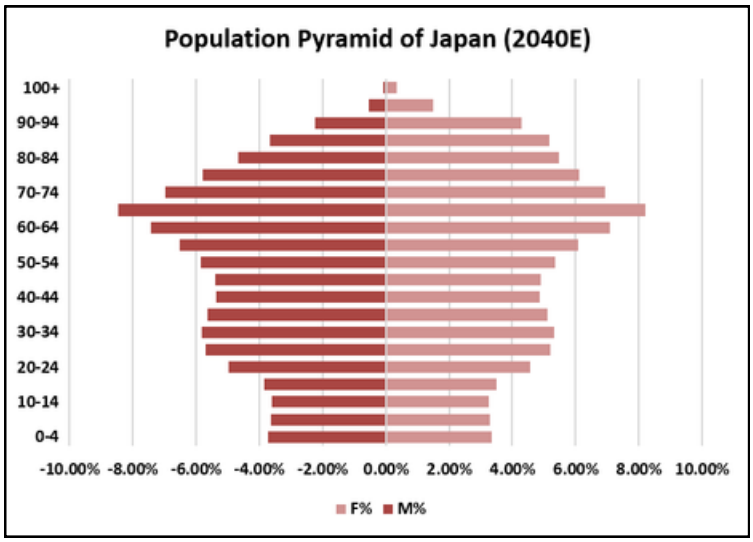
- Key aggravators: defence spending rising to **2% of GDP (¥7.5Tn),** social security growth is non-discretionary.

# Demographics and Fiscal Sustainability

***THE BANK OF JAPAN AND YIELD CURVE CONTROL***

Japan holds the highest elderly share among major economies, and its demographic structure makes the resulting fiscal pressure both severe and self-reinforcing. The pyramids show not only that Japan is old, but that the large cohorts now in their fifties and seventies migrate upward by 2040 as the working-age base narrows beneath them.




## *THE DEMOGRAPHIC ENGINE*

Japan's pension and healthcare systems run in Pay-As-You-Go **(PAYG)**: current workers fund current retirees.

As a shrinking workforce supports a growing retiree pool, contributions fall short of payouts, and the government borrows to close the gap each year. Demographics is the primary engine of the deficits behind the debt.

## *THE FISCAL FOOTPRINT*

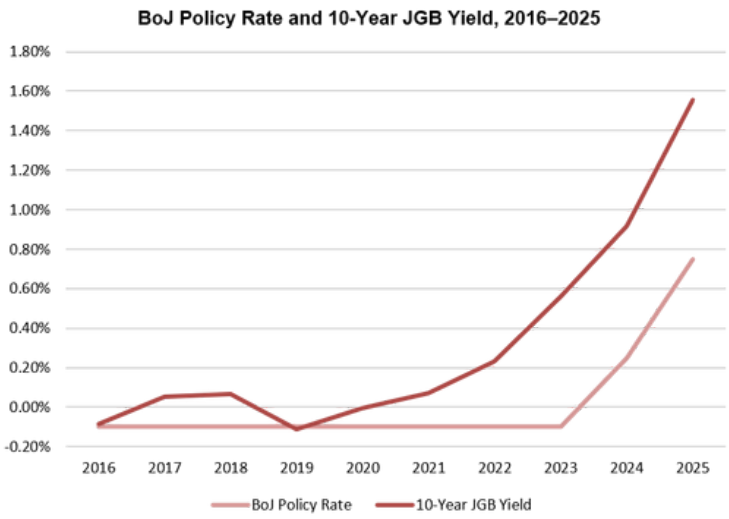
Social security benefits reached ¥135.5 trillion in FY2023 **(~22.8% of GDP)**, with pensions and healthcare making up roughly three-quarters.

Social security and debt servicing together consume close to **58% of the national budget**, compressing the room left for defence, infrastructure, and education.

- No Way to Grow Out: Aging raises mandatory spending while shrinking the tax base that funds it, tightening the squeeze from both sides automatically.
- The Worsening Trajectory: With the **65+ share heading toward ~35% by 2040.**
- Structural, Not Cyclical: This is a permanent demographic shift, not a downturn, anchoring every other challenge in Japan's debt story.

# Exit Risks and Future Scenarios

***THE BANK OF JAPAN AND YIELD CURVE CONTROL***




### *BoJ Exit Risks*

After ending negative rates and yield curve control in March 2024, the Bank of Japan has begun the most consequential exit in its history.

The policy rate has since risen from **−0.1% to 0.75%** and the 10-year yield from near **zero to above 1.5%**, and the dangers of unwinding a position this large are now becoming visible in real time.

### *Two Routes Out, Both costly*

Raising rates inflates the interest bill on a debt of **roughly 263% of GDP,** with debt-servicing spending set to top **¥30 trillion for the first time.**

Selling bonds is equally constrained: holding **49% of the JGB market**, any aggressive disposal would crash prices and spike yields.

The BoJ is therefore tapering gradually rather than selling, reducing purchases by **¥400 billion per quarter, because only an inch-by-inch exit avoids triggering the very crisis it seeks to escape.**

### *INTEREST RATE SENSITIVITY ANALYSIS*

As the BoJ steps back, the market is repricing after years of suppression, **with the 10-year yield rising faster than the policy rate as a risk premium re-emerges.**

Higher yields inflict losses on the **JGB holdings of banks, insurers, and the BoJ itself,** while feeding through to the yen and wider markets. **Each step of the exit tightens fiscal and financial conditions at once.**

The Real Signal Is the Regime Change The specific trades matter less than what they collectively mark: the end of Japan as a source of free money. For a generation, suppressed yields made JGBs uninvestable and funded global carry trades in cheap yen. Normalization reverses both, and the lasting implication is that Japan is becoming a market that once again prices risk, with consequences that extend well beyond its borders.